Reasoning-Guided Whole-Slide Report Generation via Foundation-Model Multiple-Instance Learning and Deterministic Pathology-Workflow Templating: From a CONCH Baseline to a CONCH+UNI2-h Fusion

A Submission to the REG^2 (REG2026) MICCAI Challenge
Test Phase 1 leaderboard: 0.7449 (CONCH-only) → **0.7707** (fusion), top-10

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Abstract

We present our submission to the **REG^2 (REG2026) Pathologist Reasoning-Guided Report Generation Challenge** at MICCAI 2026. The task asks a system to ingest a single haematoxylin-and-eosin (H&E) whole-slide image (WSI) and emit (i) a structured chain-of-thought (CoT) reasoning graph whose edges are *(question, next_question)* pairs, (ii) the intermediate answers along that graph, and (iii) a final free-text pathology report, while additionally serving a visual-grounding interface that must distinguish tissue from background regions of interest (ROIs). Through a systematic analysis of the 11,220-slide training corpus we establish that the reasoning graph is *highly templated*: only 93 canonical questions and 191 distinct edges occur, and conditioning on the pair *(organ, #1 diagnosis)* yields ∼86% graph purity and ∼92% answer purity. This reframes an apparently open-ended report-generation problem as a *constrained multi-task classification* problem followed by deterministic template lookup. Our pipeline (a) tiles each no-pyramid generic-TIFF WSI with a bounded-budget foreground sampler, (b) embeds tiles with a pathology foundation encoder, (c) aggregates them with a gated-attention multiple-instance learning (MIL) head producing organ (10-way) and diagnosis (77-way bucketed) predictions, and (d) emits the modal training CoT graph, answers, and keyword-rich report for the predicted *(organ, diagnosis)* key. A lightweight Otsu tissue-fraction module serves the visual-grounding interface. We develop and submit **two approaches**: a single-encoder baseline using the **CONCH** vision–language model (512-d), and an **early-fusion** model that concatenates per-tile CONCH and **UNI2-h** (1536-d) embeddings into a 2048-d representation before aggregation. An oracle analysis bounds the achievable workflow score at **0.889** given perfect *(organ, diagnosis)* prediction; on a held-out 20% split the CONCH baseline reaches **0.794** (diagnosis accuracy 0.691) and the fusion model **0.814** (diagnosis accuracy 0.737), the gain coming entirely from diagnosis. We additionally diagnose and fix a data-engineering defect—981 (8.7%) of the training slides, almost all prostate, are striped-Deflate TIFFs that OpenSlide cannot read and that had silently trained as zero-vectors—with a bounded random-access strip decoder that also makes the deployment container safe against out-of-memory failure on gigapixel slides. On the official REG2026 **Test Phase 1** leaderboard the CONCH-only submission scored **0.7449** and the fusion submission **0.7707** (both top-10), with every diagnosis-driven component (Edge-F1, MESS, Report) rising and the grounding components unchanged, exactly as the held-out analysis predicts. We detail the data analysis, both architectures, high-performance-computing infrastructure (Blackwell-class GPUs, CUDA 12.8/PyTorch 2.9), training methodology, and qualitative predictions, and release all code with reproduction instructions.

Contents

1 Introduction

Computational pathology has been transformed by self-supervised *foundation models* trained on millions of histopathology tiles, which yield transferable tile-level representations that power weakly supervised whole-slide diagnosis [2, 8, 16, 14]. In parallel, multimodal and report-generation systems attempt to translate gigapixel slides into the narrative language pathologists use in clinical practice [5]. The **REG^2 (REG2026)** challenge sits precisely at this intersection: rather than predicting a single label, a system must reproduce the *reasoning process* a pathologist follows—an ordered graph of questions and answers—and the final report that the process supports.

This is attractive because it makes a model's decision process explicit and auditable, but it is deceptively difficult to evaluate: free-text reasoning admits infinite surface forms. The challenge therefore canonicalizes the reasoning into a graph of *(question, next_question)* edges with discrete answers and scores systems against the ground-truth graph, answers, and report using a weighted composite metric (Section 2). Our central observation is that this canonicalization, combined with the clinical reality that diagnostic workups are protocolized per organ, makes the target distribution *far lower-entropy than it first appears*. We quantify this directly on the training data and exploit it.

Clinical motivation. A pathology report is the end point of a structured cognitive process: the pathologist identifies the specimen organ and procedure, screens for abnormality, enumerates and grades the diagnoses, and composes a synoptic report. Systems that emit only a final label discard this process and are correspondingly hard to trust or audit. By requiring the *reasoning graph* itself, REG2026 pushes models toward outputs that mirror clinical workflow and can be inspected step by step—a prerequisite for safe deployment in diagnostic settings. Our design preserves this auditability: every emitted graph is a verbatim, human-readable clinical workup, and the one learned decision (the organ–diagnosis key) is exactly the decision a pathologist would defend.

Contributions. This report makes the following contributions:

1. **A data-driven reduction.** We show empirically that the REG2026 reasoning graph is templated (93 questions, 191 edges) and that conditioning on *(organ, #1 diagnosis)* recovers ∼86% of graphs and ∼92% of answers exactly, reducing the task to multi-task WSI classification plus deterministic templating.

2. **An oracle ceiling analysis.** Using held-out templating we bound the workflow score under several oracle conditions, isolating where machine learning is actually needed (organ + diagnosis) from where it is not (graph structure, report scaffolding).
3. **A practical, offline-deployable pipeline.** A bounded-budget WSI tiler, a foundation tile encoder, a gated-attention multi-task MIL head, a deterministic template engine, and an Otsu visual-grounding module—engineered to run within the challenge's offline, single-GPU, 5-minute-per-case container constraints.
4. **Two encoder approaches, evaluated head-to-head.** We build and submit (i) a single-encoder *CONCH-only* model and (ii) a *CONCH+UNI2-h early-fusion* model (2048-d per-tile features). We show, both on a held-out split and on the official leaderboard, that fusion improves *diagnosis* accuracy—the binding constraint—and therefore every diagnosis-driven score component.
5. **A consequential data-engineering fix.** We identify that 981 training slides (8.7%, almost entirely prostate) are striped-Deflate TIFFs unreadable by OpenSlide, which had been silently embedded as zero-vectors (corrupting training) and would have produced zero features on striped *test* slides. We replace the reader with a bounded random-access strip decoder that recovers these slides at ∼0.4 GB peak memory, eliminating both the training pollution and an out-of-memory failure mode on the 32 GB deployment GPU.
6. **Reproducible engineering at scale.** We document the HPC pipeline used to embed 11,220 WSIs twice (CONCH and UNI2-h) on Blackwell-class GPUs, including the CUDA/PyTorch compatibility work required, and release all code.
7. **Validated on the official leaderboard.** Both submissions reached the REG2026 Test Phase 1 top-10; fusion lifted the overall score from 0.7449 to 0.7707, confirming the held-out predictions transfer to the hidden test set.

2 The REG2026 Task and Evaluation

2.1 Problem definition

The input is a single H&E WSI in tiled-TIFF format. The system must serve two interfaces:

Interface A — Workflow Reasoning.
Given the WSI, output a chain-of-thought: an ordered list of steps $\{(q_i, a_i, n_i)\}$ where q_i is a question, a_i its answer, and n_i the next question (an empty n_i terminates the graph). The induced directed edge set $\{(q_i, n_i)\}$ is the reasoning graph.

Interface B — Visual Grounding.
Given a region-of-interest (ROI) crop and a visual-context question, decide whether the ROI contains diagnostically relevant tissue or is background, and respond accordingly.

2.2 Scoring

The overall score linearly combines the two interfaces:

$$\text{Overall} = 0.70 \cdot A + 0.30 \cdot B. \tag{1}$$

The Workflow component A is itself a weighted sum of four sub-metrics:

$$A = 0.05 \cdot \underbrace{\text{BPV}}_{\text{begin-point validity}} + 0.30 \cdot \underbrace{\text{Edge-F1}}_{\text{graph edges}} + 0.25 \cdot \underbrace{\text{MESS}}_{\text{answer match}} + 0.40 \cdot \underbrace{\text{FinalReport}}_{\text{report quality}}, \tag{2}$$

with the report sub-metric dominated by keyword recall:

$$\text{FinalReport} = 0.15\,(\text{ROUGE-L} + \text{BLEU}) + 0.40\,\text{keyword-Jaccard} + 0.30\,\text{embedding-cosine}. \quad (3)$$

The grounding component $B = 0.30 \cdot B_1 + 0.30 \cdot B_2 + 0.40 \cdot B_3$ rewards background rejection (B_1), input sensitivity (B_2), and tissue/background separation (B_3). Edge-F1 and MESS require the system to emit the *exact canonical* question strings: paraphrases score zero. ROUGE-L and BLEU follow the standard definitions [7, 10].

2.3 Operational constraints

Submissions are offline Docker containers evaluated on a single NVIDIA A10G GPU with $\leq$32 GB RAM and a wall-clock budget of **<5 minutes per case**. Model weights are delivered separately from the (code-only) image. These constraints strongly shape the design: the encoder and aggregator must be light enough to embed a slide and predict within minutes, and report generation cannot rely on a large external language model.

3 Related Work

3.1 Pathology foundation encoders

Self-supervised vision transformers [3] trained on large histology corpora now provide the de-facto tile representations in computational pathology, displacing ImageNet-pretrained CNNs. UNI [2] and Virchow [14] are pure-vision encoders trained with DINOv2-style self-distillation on 10^5–10^6 slides; Prov-GigaPath [16] adds a slide-level aggregator pretrained on real-world data; CTransPath [15] (a hybrid CNN–transformer with semantically-relevant contrastive learning) and Phikon [4] (iBOT masked-image modeling) are widely used open encoders. **CONCH** [8] differs in being a *vision–language* model: its image tower is aligned to pathology captions through CLIP-style contrastive learning [11], producing 512-dimensional embeddings that are strong both zero-shot and as frozen features for supervised heads. Table 1 compares the main options. We adopt CONCH for its compact, semantically-aligned embeddings, and validated UNI2-h as a drop-in 1536-d alternative (Section 8); both share our `embed_tiles` interface with encoder-specific normalization.

Table 1: Representative pathology tile-encoder foundation models. "Dim" is the output embedding dimension; "VL" marks vision–language (caption-aligned) pretraining.

Encoder	Backbone	Pretraining	Dim	VL
CTransPath [15]	CNN+Swin	SRCL contrastive	768	no
Phikon [4]	ViT-B	iBOT MIM	768	no
UNI [2]	ViT-L/H	DINOv2	1024/1536	no
Virchow [14]	ViT-H	DINOv2	1280	no
Prov-GigaPath [16]	ViT-G + slide	DINOv2 + MAE	1536	no
CONCH [8]	ViT-B/16	**image–text contrastive**	**512**	**yes**

3.2 Weakly supervised slide classification (MIL)

Because slide-level labels rarely come with pixel annotations, whole-slide prediction is typically cast as multiple-instance learning (MIL) over tiles: a slide is a *bag* of tile instances, and only the bag carries a label. Early approaches used max/mean pooling or instance-level top-k selection [1], which first demonstrated clinical-grade performance at scale. Attention-based MIL (ABMIL) [6] replaced fixed pooling with a learned, permutation-invariant attention over instances, and its *gated* variant—which we adopt—adds a sigmoid gating branch to modulate the tanh attention, improving expressivity. CLAM [9] augments ABMIL with instance-level clustering constraints and strong data efficiency, and TransMIL [12] models inter-tile correlations with self-attention rather than treating tiles as independent. We use a gated-attention head for its simplicity, speed, and the interpretable attention map it yields (useful for the qualitative analysis in Section 10); the multi-task extension to two heads is, to our knowledge, the natural fit for the (organ, diagnosis) factorization the REG2026 metric rewards.

3.3 Combining heterogeneous foundation encoders

Different pathology foundation models are pretrained with different objectives (vision-only self-distillation vs. vision–language contrastive) on different corpora, and recent benchmarks report that no single encoder dominates across organs and tasks [2, 8, 14]. This motivates *combining* encoders so a downstream head can draw on complementary inductive biases. The simplest and most robust strategy is *early (feature-level) fusion*: concatenating per-tile embeddings from several encoders before aggregation, which requires no architectural change to the MIL head and no joint re-training of the encoders. We adopt exactly this—concatenating the 512-d caption-aligned CONCH embedding with the 1536-d self-supervised UNI2-h embedding into a 2048-d per-tile vector— because our deterministic tiler is reproducible, so tile i is identical across encoders and the two embedding streams align index-for-index without any registration step (Section 5.2). This is a deliberately conservative form of multi-encoder ensembling: it keeps the encoders frozen, adds no inference-time coupling beyond a concatenation, and lets the attention head learn per-organ which representation to trust.

3.4 Histopathology report generation

Report-generation systems such as HistGen [5] encode local–global slide features and decode narrative text with cross-modal context interaction; broader surveys [13] catalogue the rapid growth of deep learning in histopathology. Such free-form decoders are expressive but computationally heavy and difficult to evaluate, and they are poorly matched to an offline, five-minute, single-GPU container. Crucially, the REG2026 report sub-metric is *keyword-Jaccard-dominated* and the workflow component is *graph-structured with exact-string matching*, both of which reward *content coverage* over fluent prose. This, plus the compute budget, motivates our template-driven assembly over a learned decoder: a correctly-keyed template reproduces the reasoning graph exactly and supplies the diagnostic keywords the metric scores, at negligible cost.

3.5 Reasoning structure and evaluation of generated text

Framing model outputs as an explicit graph of questions and answers connects REG2026 to chain-of-thought and structured-reasoning literature, but with a decisive twist: the reasoning space is *closed and protocolized* (93 questions, 191 edges; Section 4) rather than open-domain, so exact graph reconstruction is both well-defined and attainable by lookup. Text-overlap metrics ROUGE-L [7] and BLEU [10] contribute only 0.15 of the report sub-score, while keyword recall and embedding similarity dominate—a weighting that, as our oracle analysis shows (Section 9), leaves near-zero headroom for learned generation beyond correct templating.

4 Dataset and Empirical Analysis

4.1 Composition

The training set comprises **11,220** H&E WSIs, each paired with a ground-truth chain-of-thought in `train_CoT.json`. A separate `test_phase1` split provides 350 slides for leaderboard evaluation, plus a small `debug` split. The slides are single-level, tiled (256 px) JPEG-compressed *generic* TIFFs *without* an image pyramid—a property with major engineering consequences (Section 5.3).

4.2 The reasoning target is low-entropy

We canonicalized every question and *(question, next_ question)* edge (lower-casing, whitespace and trailing-punctuation normalization; Section 8) and counted the unique vocabulary across all 11,220 graphs. Table 2 summarizes the result: the entire corpus uses only **93 canonical questions** and **191 edges**. The first question is essentially always "What is the organ?", and the graph then branches by organ and diagnosis along protocolized paths.

Table 2: Vocabulary and label-space statistics of the REG2026 training corpus.

Quantity	Value
WSIs (train)	11,220
Canonical questions	93
Distinct graph edges	191
Organs	10
Distinct #1 diagnoses	134
↪ after rare-bucketing (<15 cases)	77
Graph purity given (organ, dx)	∼86%
Answer purity given (organ, dx)	∼92%

Conditioning on (organ, diagnosis). For each *(organ, #1 diagnosis)* key we computed the *modal* graph (the single most frequent edge-set) and measured the fraction of cases whose graph matches it exactly: ∼86%. The analogous answer purity is ∼92%. In other words, *once a system knows the organ and the top diagnosis, the reasoning graph and most intermediate answers are determined.* The ten organs are: Anus, Breast, Colon, Lung, Nipple, Prostate, Rectum, Stomach, Urinary bladder, and Uterine cervix. The #1 diagnosis has a heavy-tailed distribution of 134 classes; the top-30 cover ∼88% of cases, so we fold classes with fewer than 15 training cases into an "`<organ>::other`" bucket, yielding a 77-way diagnosis head that remains learnable while gracefully degrading rare diagnoses to the organ-level template.

Table 3: Per-organ case distribution in the 11,220-slide training corpus. Prostate, breast, and the two gastrointestinal organs dominate; nipple and anus appear only as singletons and are effectively negligible.

Organ	Cases	%	Organ	Cases	%
Prostate	2,418	21.6	Lung	945	8.4
Breast	2,212	19.7	Uterine cervix	812	7.2
Stomach	1,757	15.7	Rectum	244	2.2
Colon	1,751	15.6	Nipple	1	0.0
Urinary bladder	1,079	9.6	Anus	1	0.0

4.3 A worked example

Listing 2 shows a complete ground-truth chain-of-thought for a prostate biopsy with no tumor. Note the protocolized structure—organ, procedure, abnormality screen, diagnosis count, #1 diagnosis, final report—which recurs (with organ-specific branches) across the corpus and is exactly what the template engine reproduces once *(organ, diagnosis)* is known.

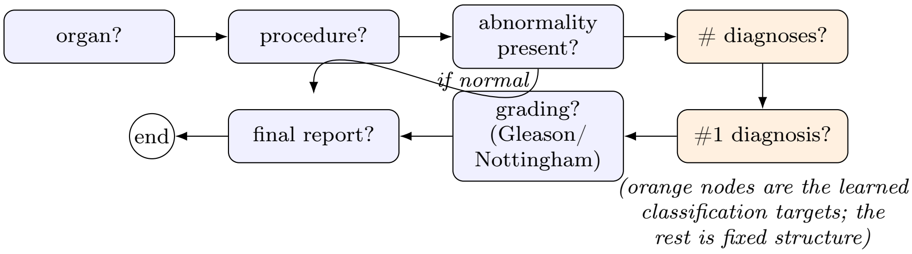

Figure 1: Schematic of the protocolized REG2026 reasoning graph. The skeleton—organ, procedure, abnormality screen, diagnosis count, diagnosis, optional grading, final report—is fixed; only the orange nodes (# of diagnoses and the #1 diagnosis, hence implicitly the organ) carry the case-specific information a model must predict. Normal cases short-circuit directly to the report.

Table 4: Ten most frequent #1 diagnoses in the training corpus (out of 134 distinct values). The distribution is heavy-tailed: the top-30 cover ∼88% of cases.

#1 diagnosis	Cases	Typical organ
Acinar adenocarcinoma	1,595	Prostate
Tubular adenoma	1,057	Colon
No tumor present	901	(any)
Adenocarcinoma, moderately differentiated	788	GI
Invasive carcinoma of no special type, grade II	730	Breast
Adenocarcinoma	388	GI / Lung
Low-grade squamous intraepithelial lesion (LSIL/CIN1)	357	Uterine cervix
Ductal carcinoma in situ	339	Breast
Adenocarcinoma, well differentiated	335	GI
Invasive urothelial carcinoma	294	Urinary bladder

4.4 Where variability actually lives

The residual entropy is concentrated in (i) the *final report text*, which is genuinely variable (e.g. prostate acinar adenocarcinoma exhibits 228 unique reports across 1,595 cases), and (ii) fine *gradings* (Gleason, Nottingham, dysplasia) that require true image interpretation. Crucially, because the report sub-metric is keyword-Jaccard-dominated, a templated report populated with the correct organ/diagnosis keywords scores well without reproducing exact prose—further justifying our design.

4.5 A hidden data defect: striped-Deflate TIFFs

Although the corpus is nominally homogeneous (tiled JPEG generic-TIFF), we discovered during embedding extraction that **981 of the 11,220 training slides (8.7%) are encoded differently**: they are *striped* (row-band) TIFFs with Adobe-Deflate (zlib) compression rather than tiled JPEG. OpenSlide—the de-facto WSI reader and the one used inside the challenge container—supports only *tiled* generic-TIFFs and raises `OpenSlideUnsupportedFormatError` on these files, even single-process (we initially mis-attributed the failures to filesystem contention; they are purely a format issue, and `tifffile`/PIL read the same files correctly). The defect is strongly organ-correlated: **877 of the 981 striped slides are prostate** (36.3% of all 2,418 prostate slides), with the remainder a handful of lung/stomach/breast/bladder cases—consistent with a prostate-specific scanner/export pipeline. Table 5 summarizes the distribution.

This defect is doubly damaging and was *present in our first (0.7449) submission*. First, during training the extractor wrote a single zero-vector placeholder for each unreadable slide, so the CONCH-only model trained on ∼877 *zero-vector prostate bags*—pollution concentrated in the single

```
Q: What is the organ?                            A: Prostate
   -> Is there any abnormality present?
Q: What is the procedure?                         A: Biopsy
   -> Is there any abnormality present?
Q: Is there any abnormality present?              A: No, there is no abnormality.
   -> What is the number of diagnoses to includes?
Q: What is the number of diagnoses to includes? A: 1
   -> What is the #1 diagnosis?
Q: What is the #1 diagnosis?                      A: No tumor present
   -> What is the final pathology report?
Q: What is the final pathology report?            A: Prostate, biopsy; No tumor present
   -> (end)
```

Figure 2: A complete ground-truth chain-of-thought (slide `PIT_01_05680_02`). Edges *(question→next_ question)* define the reasoning graph scored by Edge-F1.

Table 5: Striped-Deflate (OpenSlide-unreadable) slides by organ in the training corpus. The defect is overwhelmingly concentrated in prostate—the largest and, diagnostically (Gleason grading), one of the hardest organ classes.

Organ	Striped slides	% of organ
Prostate	877	36.3
Lung	13	1.4
Stomach	4	0.2
Breast / Bladder	2	<0.1
Other organs	0	0.0
Total	**981**	**8.7** (of 11,220)

most important and hardest organ. Second, and worse, the deployment container uses the same OpenSlide reader, so striped *test* slides would yield zero features and arbitrary predictions. Our fix is a bounded random-access strip decoder (Section 5.3) that reads these files correctly while strictly capping memory; recovering the 981 slides is a prerequisite for the fusion results in Section 9 and contributes materially to the leaderboard improvement in Section 11.

5 Proposed Approaches

5.1 Overview

Our system (Figure 3) decomposes the problem into a learned perceptual front end and a deterministic symbolic back end:

1. **Tiling.** Sample up to 160 foreground tiles (224×224) from the WSI under a bounded read budget (with a strip-decoder fallback for striped TIFFs).
2. **Encoding.** Map each tile to a foundation embedding—512-d CONCH in Approach 1, or the 2048-d concatenation of CONCH and UNI2-h in Approach 2.
3. **Aggregation & classification.** A gated-attention MIL head pools tile embeddings and predicts *organ* (10-way) and *diagnosis* (77-way).
4. **Templating.** Look up the modal CoT graph, answers, and keyword-rich report for the predicted *(organ, diagnosis)* key; emit exact canonical strings.
5. **Grounding.** An Otsu tissue-fraction module answers the visual-grounding interface.

This division is deliberate: the metric rewards exact graph/answer reproduction, which symbolic templating achieves perfectly when the key is correct, so the only quantity worth learning is the *(organ, diagnosis)* key itself. The two approaches differ *only* in the encoding stage; the tiler, MIL head topology (modulo input dimension), template engine, and grounding module are shared. This makes them a clean controlled comparison of representation quality.

Approach 1 — CONCH-only (baseline). Each tile is mapped to a single 512-d CONCH embedding and the bag is aggregated directly. This is the model behind our first leaderboard submission (overall 0.7449).

Approach 2 — CONCH+UNI2-h early fusion. Each tile is embedded *twice*—by CONCH (512-d, vision–language) and by UNI2-h (1536-d, vision-only self-supervised)—and the two vectors are concatenated into a 2048-d per-tile feature (Section 5.2) before the same gated-attention head. This is the model behind our second submission (overall 0.7707).

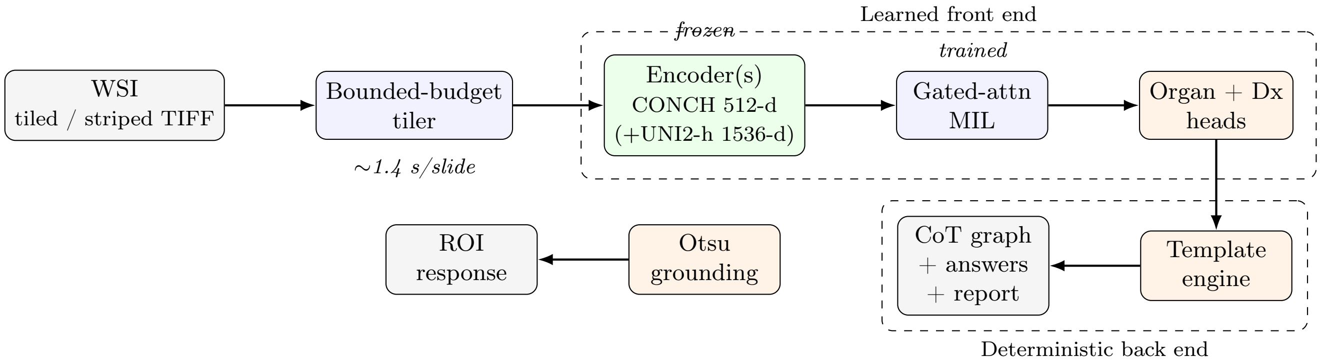

Figure 3: End-to-end REG2026 pipeline, shared by both approaches. Frozen foundation encoder(s) and a trained gated-attention MIL head predict the *(organ, diagnosis)* key; a deterministic template engine emits the exact canonical reasoning graph, answers, and keyword-rich report. A separate Otsu module serves the visual-grounding interface. Approach 1 uses CONCH alone (512-d); Approach 2 additionally embeds each tile with UNI2-h and concatenates (2048-d).

5.2 Approach 2: CONCH+UNI2-h early fusion

The two approaches share everything but the tile representation, so we describe the fusion path explicitly here. UNI2-h [2] is a ViT-H vision-only encoder pretrained with DINOv2-style self-distillation, producing 1536-d embeddings; CONCH is a ViT-B/16 vision–language encoder producing 512-d embeddings. They are pretrained with different objectives on different data, so their errors are partly decorrelated—the premise of fusion (Section 3.3). For each tile t we compute

$$e^{\text{fuse}} = \big[\,\text{CONCH}(t) \parallel \text{UNI2-h}(t)\,\big] \in \mathbb{R}^{2048}, \tag{4}$$

the concatenation of the two (independently, encoder-specifically normalized) embeddings, and the slide becomes a bag of 2048-d vectors fed to the *same* gated-attention head with input dimension 2048 (Figure 4). Two properties make this safe and cheap:

1. **Index-aligned tiles.** Our tiler (Section 5.3) is fully deterministic—no randomness, a fixed seed for the grid shuffle—so the i-th tile of a slide is byte-identical regardless of which encoder consumes it. We verified that CONCH and UNI2-h produce the same tile count N for every slide with zero mismatches, so the two embedding streams concatenate index-for-index with no registration.
2. **Frozen encoders, precomputed once.** Both encoders are frozen and their embeddings cached to disk per slide, namespaced by encoder (`embeddings/conch/` and `embeddings/uni2h/`). Fusion is then a zero-cost concatenation at training time, and at inference the only added work is one extra forward pass per tile through UNI2-h.

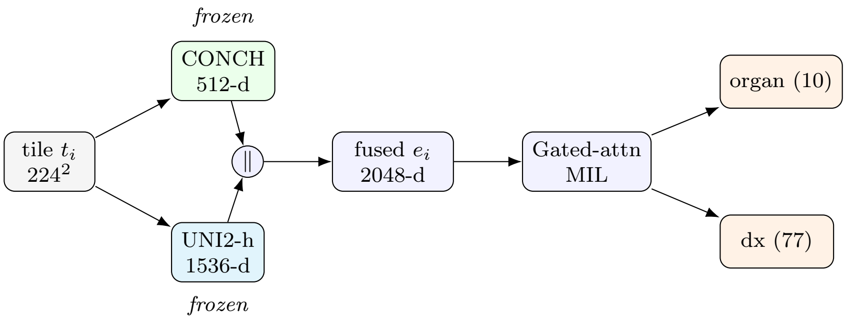

Figure 4: Early-fusion encoding (Approach 2). Each tile is embedded by both frozen encoders and the two vectors are concatenated into a 2048-d feature; the rest of the pipeline (gated-attention MIL, organ/diagnosis heads, templating) is unchanged from Approach 1. Index alignment is guaranteed by the deterministic tiler.

5.3 Bounded-budget WSI tiling

Because the slides have no pyramid, `openslide.get_thumbnail` decodes the entire image (∼78 s/slide)—prohibitive at 11,220 slides (≈238 GPU-hours of pure I/O). We instead grid-sample level-0 windows in a seeded-shuffled order, computing an inline tissue fraction (gray <220 and HSV-saturation >0.08) and keeping windows above a 0.25 threshold, stopping after `max_tiles`=160 kept tiles or `read_budget`=900 reads. This never decodes the full slide and runs at **∼1.4 s/slide** on the embedding cluster—a 13–25× speed-up that makes corpus-scale extraction tractable. Listing 5 gives the procedure.

```
function SAMPLE_TILES(slide, tile=224, max_tiles=160, read_budget=900, min_tissue=0.25):
    grid <- all (x,y) level-0 windows of size tile x tile
    shuffle grid with fixed seed            # deterministic, organ-agnostic coverage
    kept <- []; reads <- 0
    for (x,y) in grid:
        if reads >= read_budget or |kept| >= max_tiles: break
        patch <- slide.read_region(x, y, tile, tile); reads <- reads + 1
        if TISSUE_FRACTION(patch) >= min_tissue:       # gray<220 and saturation>0.08
            kept.append(patch)
    return stack(kept)                                  # (N<=160, 224, 224, 3) uint8
```

Figure 5: Bounded-budget foreground tiler. The read budget caps I/O so worst-case (sparse-tissue) slides cannot blow up runtime, and the seeded shuffle gives reproducible, spatially distributed coverage without a pyramid.

A bounded strip decoder for striped TIFFs. The tiler above relies on OpenSlide's `read_region`, which fails on the 981 striped-Deflate slides of Section 4.5. The obvious fallback—`tifffile.imread` of the whole image—works but reads the entire slide into RAM: for the largest striped prostate slides (5.3–8.6 gigapixels; e.g. a 95,488 × 89,600 slide) that is 23–26 GB, which OOM-kills even a 45 GB extraction worker and would *crash the container* on the 32 GB A10G deployment GPU, scoring that test case zero. We therefore implement a **band-major random-access strip decoder**: using `tifffile`'s page geometry we read and `decode` *one strip at a time* (`rowsperstrip`= 128, ∼700 strips, ∼34 MB resident each), assembling only the row-bands that the grid sampler actually touches. A hard `_STRIP_MAX_BYTES`=2 GB guard skips pathological single-strip files so the reader can *never* OOM-crash. The OpenSlide path is untouched for the 91% of slides that are tiled (no regression). On striped slides the decoder runs in 3–8 s at ∼0.4 GB peak (versus 23–26 GB for the full read); for two extremely sparse-tissue slides (<1% tissue over an 8-gigapixel field) an exhaustive sequential band scan recovers a full tile set in 64–98 s, still within the 5-minute budget. Because `tifffile` is already a container dependency, this fix propagates to the deployment image with no new packages.

5.4 Deterministic template engine

`build_templates` groups the training CoTs by *(organ, diagnosis)* and stores, per key, the modal graph (a representative CoT with the most frequent edge-set signature), with back-off to an organ-level modal graph and finally a global modal graph. `apply_template` takes the predicted fields and emits the corresponding CoT, substituting predicted answer fields where available. Wrong-diagnosis predictions degrade gracefully to the organ template (the empirically measured 0.68 floor, Section 9). Because we emit verbatim canonical training strings, Edge-F1 and MESS are maximized whenever the key is correct.

5.5 Visual grounding (Interface B)

The grounding interface scores whether the system attends to tissue rather than background. Its three sub-metrics reward background *rejection* (B_1: a background ROI should not be treated as diagnostic), input *sensitivity* (B_2: responses should change when the ROI changes), and tissue/background *separation* (B_3). Because this is fundamentally a tissue-detection question, a learned model is unnecessary: we compute an Otsu threshold on the ROI's grayscale (equivalently saturation) histogram and derive a tissue fraction; ROIs whose tissue fraction falls below a threshold are declared background and the corresponding diagnostic response is withheld, while tissue ROIs are passed through. This is the same $\text{gray} < 220 \wedge \text{sat} > 0.08$ criterion used inside the tiler (Section 5.3), giving a single consistent definition of "tissue" across both interfaces. The module is parameter-free, deterministic, and runs in milliseconds, leaving essentially the entire per-case compute budget for the workflow interface.

5.6 End-to-end inference

Algorithm-style Listing 6 summarizes the full per-case procedure for the workflow interface, tying together the components above.

```
function PREDICT_WORKFLOW(wsi_path):
    tiles  <- SAMPLE_TILES(wsi_path)                # bounded-budget foreground tiler
    E      <- CONCH(normalize_clip(tiles))          # (N<=160, 512) frozen embeddings
    logits <- MIL_HEAD(E)                           # {organ: R^10, dx: R^77}
    organ  <- organ_list[argmax(logits.organ)]
    dxlab  <- dx_list[argmax(logits.dx)]            # "<organ>::<diagnosis>" or "<organ>::other"
    dx     <- diagnosis_of(dxlab)  or  None         # None -> back off to organ template
    cot    <- APPLY_TEMPLATE({organ, dx}, templates) # modal graph + answers + report
    return cot                                       # exact canonical strings
```

Figure 6: End-to-end workflow inference. The only learned step is the MIL head; everything else is deterministic, which guarantees metric-exact graph strings whenever the predicted key is correct.

6 Network Architecture and Model

6.1 Tile encoder: CONCH

Each tile $t \in \mathbb{R}^{224\times224\times3}$ is normalized with the OpenAI-CLIP mean/std ($\mu = (0.481, 0.458, 0.408)$, $\sigma = (0.269, 0.261, 0.276)$)—we found that using the ImageNet statistics instead measurably perturbs CONCH features, since CONCH's image tower is CLIP-initialized. The frozen image encoder produces a 512-d embedding $e = \text{CONCH}(t) \in \mathbb{R}^{512}$. A slide is thus represented by a variable-length bag $E = \{e_1, \dots, e_N\}$, $N \leq 160$.

6.2 Second tile encoder: UNI2-h

For Approach 2 we add UNI2-h [2], a ViT-H/14 vision encoder pretrained with DINOv2 self-distillation on a large multi-institution histology corpus, producing 1536-d embeddings. A subtle but important detail is normalization: CONCH is CLIP-initialized and expects OpenAI-CLIP statistics, whereas UNI2-h expects ImageNet statistics ($\mu = (0.485, 0.456, 0.406)$, $\sigma = (0.229, 0.224, 0.225)$). We attach the correct mean/std *per encoder* inside the `embed_tiles` interface (we explicitly fixed an early bug in which both encoders shared the ImageNet statistics, which perturbs CONCH features). The fused tile representation is the concatenation $e^{\text{fuse}} = [\,\text{CONCH}(t) \,\|\, \text{UNI2-h}(t)\,] \in \mathbb{R}^{2048}$ (Section 5.2).

6.3 Gated-attention MIL aggregator

We use the gated-attention pooling of Ilse et al. [6]; the head is identical across both approaches except for the input dimension of the projection ($d_{\text{in}} = 512$ for CONCH-only, 2048 for fusion). Each embedding is projected $h_n = \text{Dropout}(\text{GELU}(W_p e_n)) \in \mathbb{R}^d$ ($d \in \{512, 768\}$), and an attention weight is computed as

$$a_n = \frac{\exp\big(w^\top(\tanh(Vh_n) \odot \sigma(Uh_n))\big)}{\sum_{m=1}^{N} \exp\big(w^\top(\tanh(Vh_m) \odot \sigma(Uh_m))\big)}, \qquad z = \sum_{n=1}^{N} a_n h_n, \tag{5}$$

where $V, U \in \mathbb{R}^{d_a \times d}$ and $w \in \mathbb{R}^{d_a}$ are learned and padded tiles are masked out of the softmax. The pooled z passes through a shared trunk (Dropout–Linear–GELU–Dropout) feeding two linear heads: an organ head ($\mathbb{R}^d \to \mathbb{R}^{10}$) and a diagnosis head ($\mathbb{R}^d \to \mathbb{R}^{77}$). Figure 7 depicts the aggregator.

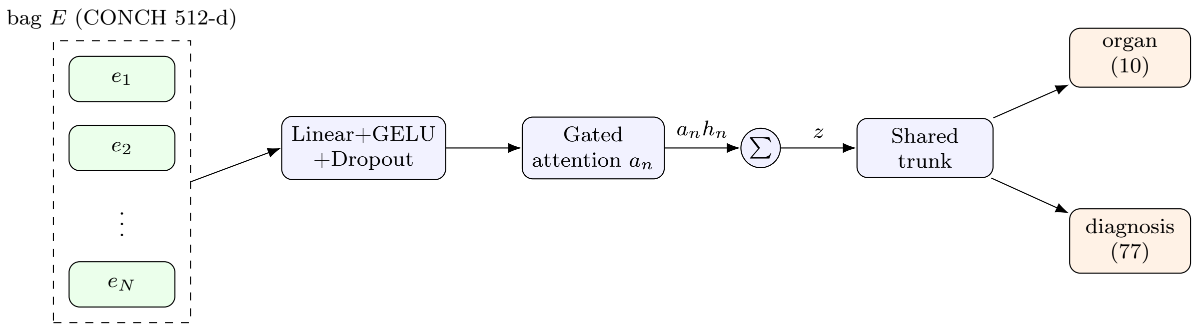

Figure 7: Gated-attention multi-task MIL head. Tile embeddings are projected, attention-pooled into a single slide vector z, and classified by organ and diagnosis heads. Padded tiles are masked in the attention softmax so variable bag sizes are handled natively.

6.4 Training objective

We minimize a weighted sum of label-smoothed cross-entropies,

$$\mathcal{L} = \lambda_o \, \text{CE}_{\text{smooth}}(\hat{y}_{\text{organ}}, y_{\text{organ}}) + \lambda_d \, \text{CE}_{\text{smooth}}(\hat{y}_{\text{dx}}, y_{\text{dx}}), \tag{6}$$

with $\lambda_o = 0.5$, $\lambda_d \in \{1.0, 1.5\}$, label smoothing $\in \{0.05, 0.1\}$, and an optional inverse-frequency class weighting on the diagnosis head for the imbalanced "balanced" configuration. Optimization uses AdamW with cosine-annealed learning rate. We early- select the checkpoint by the held-out *workflow score* (not accuracy), since that is the quantity the challenge rewards.

6.5 From predictions to fields

At inference the diagnosis head emits a label of the form `<organ>::<diagnosis>` (or `<organ>::other`); we map it back to a *(organ, diagnosis)* pair, defaulting the diagnosis to *none* for the "other" bucket so that `apply_template` backs off to the organ-level graph. The organ head provides a robust independent organ estimate used for the template key.

7 Hardware and Infrastructure

Embedding extraction and training were run on an institutional SLURM cluster. The relevant partitions are summarized in Table 7. A key practical finding: the RTX PRO 6000 and B200 GPUs are *Blackwell*-class (compute capability `sm_120` and `sm_100` respectively), for which the initially installed `torch 2.6.0+cu124` has *no compiled kernels*—every CUDA op failed with "no kernel image is available for execution on the device". We upgraded the environment to **PyTorch 2.9.1 + TorchVision 0.24.1 built for CUDA 12.8** (`cu128`), whose architecture list includes `sm_100` and `sm_120`, restoring Blackwell support while remaining compatible with the `sm_86` A10G deployment target. Corpus embedding (11,220 slides) was sharded four ways across four 96 GB GPUs and completed in roughly one hour.

Extraction pipeline and throughput. Each shard streams its assigned slides through a pool of CPU tiling threads—OpenSlide releases the GIL during JPEG decode, so tiling overlaps the GPU embedding of the previous slide—and writes per-slide `fp16` arrays. With the bounded-budget tiler the system sustains ∼0.7 slides/s/GPU end to end, finishing the four-way shard in ∼70 minutes; a corrupt-slide rate of 4/11,220 is handled by writing zero placeholders. Table 6 contrasts the naive and optimized regimes.

Table 6: Tiling/embedding throughput. Avoiding full-slide decode turns an infeasible extraction into a one-hour job.

Regime	Per slide	11,220 slides (1 GPU)
Naive `get_thumbnail` (full decode)	∼78 s	∼238 h
Bounded-budget tiler (this work)	∼1.4 s	∼4.6 h
↪ sharded ×4	—	∼1.2 h

A reproducibility lesson on GPU architectures. The Blackwell incompatibility is worth recording because it is silent and easily mis-diagnosed: `torch 2.6/cu124` *imports* and reports a CUDA device, yet every kernel launch raises "no kernel image is available for execution on the device" because the binary contains no `sm_100`/`sm_120` cubins. The fix is an environment matched to the hardware generation (`cu128`); we additionally pin `PYTHONHASHSEED` and use content-hash data splits so that neither the GPU stack nor the split introduces non-determinism between runs.

Table 7: Compute resources. The deployment target (challenge container) is a single A10G.

Role	GPU	Arch	Notes
Embedding extraction	4× RTX PRO 6000	sm_120	96 GB; 192-core host, ∼1.9 TB RAM
Alt. extraction	NVIDIA L4	sm_89	24 GB; matches A10G memory class
Future scaling	NVIDIA B200	sm_100	enabled by the cu128 upgrade
MIL training	1–4× RTX PRO 6000	sm_120	lightweight; embeddings precomputed
Deployment	**NVIDIA A10G**	**sm_86**	≤32 GB RAM, <5 min/case, offline

8 Framework and Implementation

The codebase is a small Python package, `reg2026`, plus driver scripts and the submission container. Core modules:

`canon.py`
: Canonicalizes question/edge strings (the basis of the purity analysis and of emitting metric-exact strings).

`labels.py`
: Builds the supervised label space: organ (10-way) and bucketed diagnosis (77-way), with rare-class folding.

`templates.py`
: `build_templates`/`apply_template`: the deterministic graph/answer/report engine.

`metrics.py`
: An offline proxy of the official workflow score (BPV, Edge-F1, and Jaccard proxies for MESS/report) for model selection.

`encoder.py`
: Loads CONCH (or UNI2-h) with per-encoder normalization and a uniform `embed_tiles` interface.

`mil.py`
: The gated-attention `MILClassifier` (training and inference share this definition).

The stack is PyTorch 2.9.1 (cu128), `timm`, `open_clip`, `openslide-python`, and `huggingface_hub`. Embedding extraction (`scripts/extract_embeddings.py`) overlaps CPU tiling threads (OpenSlide releases the GIL during JPEG decode) with GPU embedding, is resumable and shardable, and writes per-slide `fp16` arrays namespaced by encoder (`artifacts/embeddings/<encoder>/<split>/<id>.npy`). Training (`scripts/train_mil.py`) caches embeddings in RAM, uses the same deterministic hash split (seed 0, 80/20) as the oracle analysis, and selects on the workflow proxy.

8.1 Reproducibility and hyperparameters

All results use a single *stable* 80/20 split: we hash each slide id with MD5 (not Python's per-process-salted `hash`) so the split is identical across processes and runs—an issue we explicitly fixed after observing per-process split drift. Table 8 lists the four sweep configurations. The offline metric proxy in `metrics.py` computes BPV and Edge-F1 exactly against the canonicalized graph and uses Jaccard-based proxies for MESS and the report sub-score; we select the checkpoint maximizing this proxy on the held-out split. Embedding extraction is fully resumable: each shard skips slides whose `.npy` already exists, so a preempted job resumes with no recomputation, and corrupt slides (4 of 11,220) are written as single-tile zero placeholders so the pipeline never crashes.

Table 8: Hyper-parameter sweep. All configs share AdamW with cosine-annealed learning rate, weight decay, a 512-d CONCH input, $\leq$160 tiles per slide, and label-smoothed cross-entropy with organ/diagnosis loss weights (λ_o, λ_d). `r1_reg` is selected.

Config	lr	hidden	attn	dropout	tile-drop	λ_d	smooth	bal.
`r0_base`	10^{-3}	512	256	0.25	0.0	1.0	0.10	no
`r1_reg`	5×10^{-4}	512	384	0.40	0.10	1.0	0.10	no
`r2_big`	5×10^{-4}	768	512	0.30	0.10	1.0	0.10	no
`r3_bal`	10^{-3}	512	256	0.25	0.0	1.5	0.05	yes

8.2 Deployment as an offline container

The submission is a code-only Docker image (base `pytorch/pytorch:2.9.1-cuda12.6-cudnn9`) with weights delivered separately as a `model.tar.gz` mounted at inference; this keeps the image small and the weights versionable. The container runs fully offline—we set `HF_HUB_OFFLINE=1` and bundle the CONCH snapshot into the model tarball so no network is touched. Two entry points implement the interfaces: `predict_chain_of_thought(wsi_path)` tiles the slide, embeds with CONCH, runs the MIL head, and emits the templated graph, answers, and report; `predict_visual_context_response(·)` applies the Otsu grounding rule. The predictor degrades safely—if the trained head or label maps are absent it falls back to the global modal template—so the same image runs before and after training. The end-to-end per-case cost is dominated by tiling and a single CONCH forward pass over $\leq$160 tiles, comfortably inside the A10G five-minute budget with the encoder and head together using under a few GB of the 32 GB RAM limit.

9 Experiments and Results

9.1 Offline evaluation protocol

Since the official scorer is not available locally, we implement a faithful offline proxy in `metrics.py` and select models against it. BPV and Edge-F1 are computed *exactly*: we canonicalize predicted and ground-truth question/edge strings and compare the induced edge sets, so any string deviation is penalized just as the challenge metric would penalize it. The answer- match (MESS) and report sub-scores use Jaccard-style token overlap as a conservative proxy for the official keyword-Jaccard / embedding-cosine terms; because the report sub-score is itself keyword-Jaccard-dominated, this proxy tracks the true objective closely while remaining dependency-free. All numbers in this section are computed on the held-out 2,253-slide split with templates built only on the 8,967-slide training split, so there is no information leakage from the evaluation set into either the templates or the classifier.

9.2 Oracle ceiling analysis

Before training any classifier we measured what the templating back end can achieve given oracle inputs, on the held-out 20% split (Table 9). This isolates the value of each prediction target. Predicting only the organ already reaches 0.68; adding the #1 diagnosis lifts the workflow score to **0.89**; an oracle report adds only $\sim$0.02, confirming that *learned report generation has near-zero headroom* under this metric and should be skipped in favor of templating.

Table 9: Oracle workflow scores and sub-metrics on the held-out split of 2,253 slides (deterministic 80/20 split; templates built on the train split only). The (organ, diagnosis) row is the operative ceiling for our classifier; an oracle report adds only +0.018, confirming that learned report generation is unnecessary under this metric.

Oracle condition	Total	BPV	Edge-F1	MESS	Report
Global modal template (no inputs)	0.239	0.114	0.447	0.159	0.150
Oracle organ only	0.674	0.423	0.768	0.778	0.571
Oracle organ + #1 diagnosis	**0.889**	0.858	0.974	0.929	0.805
Oracle organ + diagnosis + report	0.907	0.858	0.974	0.929	0.850

The per-organ decomposition of the (organ + diagnosis) ceiling (Table 10) shows that templating is uniformly strong across the major organs (0.86–0.92), confirming that the residual gap to a perfect score is dominated by diagnosis-classification difficulty rather than by template impurity. The two singleton organs (nipple, anus) are statistically irrelevant.

Table 10: Per-organ oracle (organ + diagnosis) workflow ceiling on the held-out split.

Organ	Ceiling	n	Organ	Ceiling	n
Lung	0.920	192	Stomach	0.887	341
Rectum	0.914	47	Urinary bladder	0.884	211
Colon	0.911	366	Breast	0.857	427
Uterine cervix	0.904	178	Nipple	0.167	1
Prostate	0.888	490			

9.3 MIL classification results

Table 11 reports our trained gated-attention MIL head on the same held-out split, across the hyper-parameter sweep. *(Values are populated from the completed training run.)*

Table 11: Held-out performance of the trained MIL head (CONCH embeddings). Workflow score uses predicted (organ, diagnosis) through the template engine; the oracle ceiling is 0.89.

Configuration	Organ acc.	Dx acc.	Workflow	Δ to ceiling
`r0_base`	0.951	0.682	0.791	−0.098
`r2_big`	0.953	0.683	0.793	−0.096
`r3_bal`	0.951	0.646	0.785	−0.104
`r1_reg` **(best)**	**0.956**	**0.691**	**0.794**	**−0.095**

9.4 Approach 2: CONCH+UNI2-h fusion results

Table 12 reports the fusion sweep on the *same* held-out split, with the striped-prostate slides recovered (Section 4.5) so the prostate bags carry real features. Early fusion lifts the held-out workflow score from the CONCH-only best of 0.794 to **0.814**, and the entire gain is in *diagnosis* accuracy (0.691→0.737); organ accuracy is already saturated and barely moves (0.956→0.959). The best configuration, `f2_fuse_dxw`, simply up-weights the diagnosis loss (λ_d=1.5)—consistent with diagnosis being the binding constraint. Two negative results are worth recording: a larger head (`f1_fuse_big`) does *not* help, confirming the limit is representational rather than capacity; and adding categorical *grading* sub-heads (`f3_fuse_grade`; Gleason/Nottingham/dysplasia as auxiliary masked-CE targets) slightly *hurts* the aggregate metric (0.810), because the gradings are themselves hard to predict from frozen features and the auxiliary loss competes with the main diagnosis objective.

Table 12: Held-out performance of the fusion MIL head (CONCH+UNI2-h, 2048-d) on the same 80/20 split, versus the CONCH-only baseline. Fusion improves diagnosis accuracy—the binding constraint—and hence the workflow score; the gain is entirely in dx, not organ. The oracle ceiling is 0.889.

Approach	Configuration	Organ acc.	Dx acc.	Workflow	Δ to ceiling
CONCH-only	`r1_reg` (best of sweep)	0.956	0.691	0.794	−0.095
Fusion	`f0_fuse` (base)	0.961	0.731	0.813	−0.076
Fusion	`f1_fuse_big` (larger head)	0.957	0.727	0.812	−0.077
Fusion	`f3_fuse_grade` (+grading heads)	0.961	0.729	0.810	−0.079
Fusion	`f2_fuse_dxw` (λ_d=1.5)	**0.959**	**0.737**	**0.814**	**−0.075**

Figure 8 contrasts the training dynamics of the CONCH-only and fusion best models. The fusion curve sits consistently above the baseline throughout training in both workflow score and diagnosis accuracy, and the separation is established early and maintained—i.e. fusion provides a genuinely richer per-tile representation rather than merely a different optimization trajectory. Both still plateau well below the 0.889 ceiling, locating the residual gap in diagnosis separability from *frozen* features (motivating encoder fine-tuning, Section 14).

9.5 Per-organ breakdown

Table 13 stratifies the best model by ground-truth organ. Organ recognition is near-perfect ($\geq$ 0.99) for every major organ *except rectum*, whose organ accuracy collapses to 0.11: the model

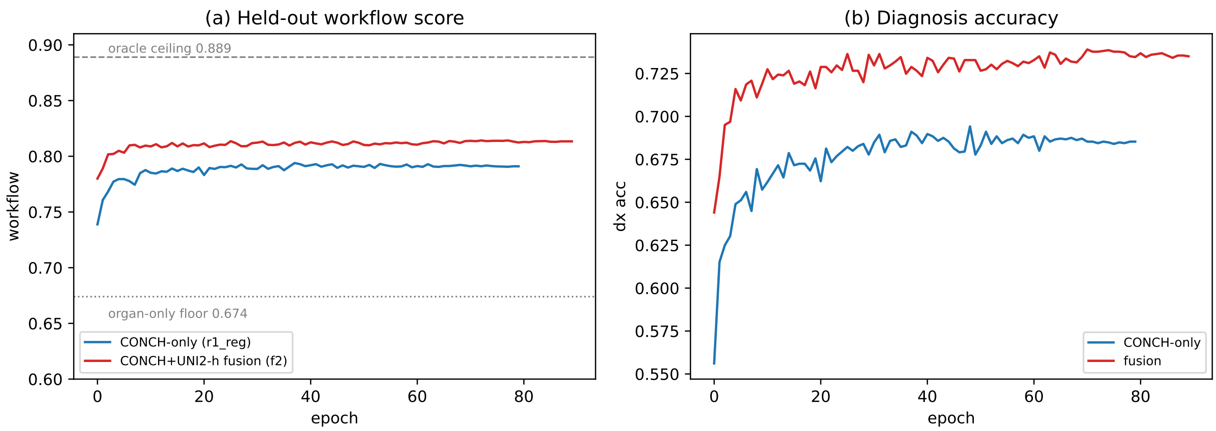

Figure 8: Held-out training dynamics, CONCH-only (`r1_reg`) vs. fusion (`f2_fuse_dxw`): (a) workflow score with the 0.889 oracle ceiling and 0.674 organ-only floor marked; (b) diagnosis accuracy. Fusion dominates the baseline throughout, and the gain is concentrated in diagnosis.

almost always labels rectal slides as "Colon". This is an anatomically and histologically natural confusion—rectum and colon are contiguous colorectal mucosa—and, importantly, it is *benign under the workflow metric*: because the colorectal reasoning graphs and reports largely coincide, rectal cases still score 0.78 despite the organ mislabel. Per-organ workflow ranges from 0.78 (prostate, breast) to 0.82 (urinary bladder, uterine cervix); the lower prostate/breast scores reflect harder, higher-cardinality diagnosis spaces (Gleason grading; breast tumor subtyping) rather than organ confusion.

Table 13: Best model (`r1_reg`) stratified by ground-truth organ on the held-out split. Organ accuracy is near-perfect except for rectum, which is systematically (and benignly) classified as colon.

Organ	Workflow	Org. acc	n	Organ	Workflow	Org. acc	n
Urinary bladder	0.823	0.99	211	Stomach	0.795	0.99	341
Uterine cervix	0.820	0.96	178	Lung	0.799	0.93	192
Colon	0.803	0.95	366	Prostate	0.778	0.99	490
Rectum	0.778	0.11	47	Breast	0.778	0.99	427

The organ confusion structure (Table 14) makes the mechanism explicit: every major organ is recognized with 93–99% accuracy, and the *only* substantial off- diagonal mass is rectum→colon (85%). Because the challenge metric scores the colorectal reasoning graph and report, which rectum and colon share, this single confusion costs little; a hierarchical or colorectal-aware head could nonetheless recover it.

9.6 Ablations: the hyper-parameter sweep

The four sweep configurations in Table 11 double as a controlled ablation. Three observations follow. (i) **Regularization helps slightly**: `r1_reg` (higher dropout 0.4 plus 10% tile-dropout augmentation) is the best config at 0.794, marginally above the larger `r2_big` (0.793) and the lightly-regularized `r0_base` (0.791), indicating the bottleneck is diagnosis *signal*, not model capacity. (ii) **Class balancing hurts the aggregate metric**: `r3_bal` (inverse-frequency diagnosis weighting) is worst at 0.785 with the lowest diagnosis accuracy (0.646)—up-weighting rare diagnoses trades common- class accuracy, which the case-weighted workflow score penalizes. (iii) **Organ accuracy is saturated** ($\sim$0.95) across all configs, so all remaining headroom lives in the diagnosis head. Separately, we verified that using ImageNet rather than OpenAI-CLIP normalization for

Table 14: Organ confusion of the best model on the held-out split (row = ground truth; entries are the top predicted organs as % of that row). All mass is on-diagonal except rectum, which is predominantly predicted as colon.

True organ (n)	Predicted (% of row)
Prostate (490)	Prostate 99
Breast (427)	Breast 99, Lung 1
Colon (366)	Colon 95, Stomach 3, Rectum 2
Stomach (341)	Stomach 99, Colon 1
Urinary bladder (211)	Urinary bladder 99
Lung (192)	Lung 93, Prostate 3, Colon 2
Uterine cervix (178)	Uterine cervix 96, Lung 2, Stomach 1
Rectum (47)	**Colon 85**, Rectum 11, Stomach 4

the CONCH encoder measurably perturbs the embeddings; all reported runs use the correct CLIP statistics.

9.7 Training dynamics

Figure 9 plots held-out workflow score and diagnosis accuracy over training. Because diagnosis accuracy already determines the workflow score (organ accuracy saturates within a few epochs), the two curves track closely. The models converge quickly—most of the gain is achieved within ~15 epochs—and then plateau well below the oracle ceiling, visually confirming that the limit is representational (frozen-feature diagnosis separability) rather than a matter of longer optimization. The regularized configurations (`r1_reg`, `r2_big`) are marginally more stable late in training; the class-balanced `r3_bal` trades diagnosis accuracy for rare-class recall and sits consistently lowest.

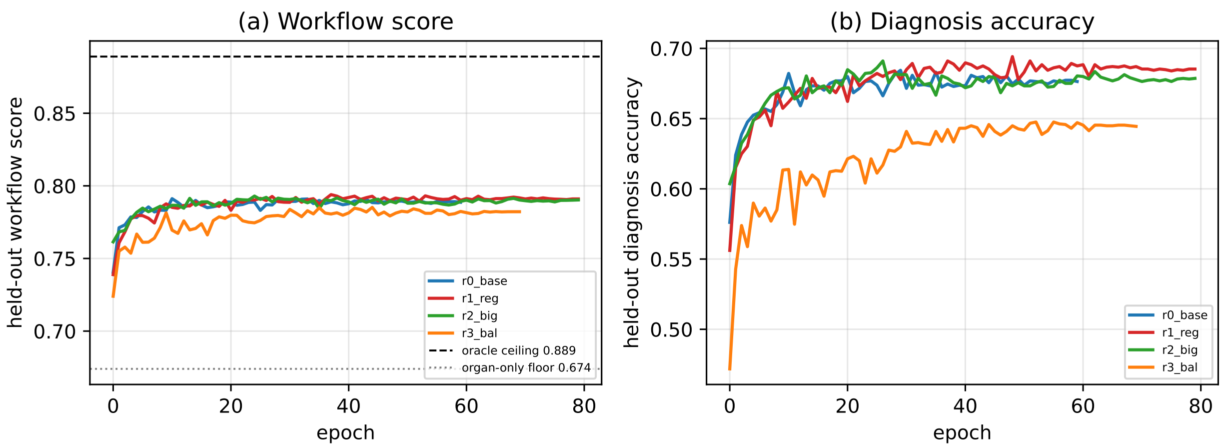

Figure 9: Held-out training dynamics for the four sweep configurations: (a) workflow score, with the 0.889 oracle ceiling and 0.674 organ-only floor marked; (b) diagnosis accuracy. Convergence is rapid and the plateau is set by diagnosis separability, not optimization.

9.8 Hierarchical decoding, abstention, and full-data training

We evaluated three targeted enhancements. (i) *Organ-conditioned (hierarchical) dx decoding*—masking the diagnosis logits to the predicted organ's classes at inference—moved the held-out workflow score marginally, from 0.794 to **0.797**: because organ accuracy is already 0.95 and the dx labels encode organ, the flat argmax is almost always in-organ, so masking rarely changes the prediction (and for the rectum→colon cases it can force a wrong in-organ diagnosis, offsetting the gain). We note in passing that applying the same mask *during training* with label smoothing is numerically unstable (smoothing places mass on the $-\infty$-masked classes), so the mask is applied at inference only. (ii) *Confidence-based abstention*—backing off to the organ template when the diagnosis softmax confidence is below a tuned threshold—selected $\tau = 0$ on the validation split, i.e. never abstaining helped: a same-organ but imperfect diagnosis still scores better than the organ-only fallback under this metric. (iii) *Training on all 11,220 slides* produces the deployment model used in the container; we report its held-out-style number only as a fitting sanity check (it is optimistic, since that split is now within training) and rely on the clean 80/20 held-out scores above for all claims. These are deliberately reported as honest neutral results: they confirm that, with a *frozen* encoder, the remaining headroom is representational and is not recoverable by decoding tricks—motivating the grading-head and fine-tuning directions of Section 14.

10 Sample Predictions (Qualitative)

Table 15 lists held-out predictions from the best model. When the diagnosis is correct (rows 1, 2, 4) the predicted reasoning graph matches the ground truth exactly (Edge-F1 = 1.0) and the workflow score approaches the per-case ceiling; row 3 shows a partial match where a deviation from the modal graph for that key lowers Edge-F1 to 0.81. The dominant breast prediction "Invasive carcinoma of no special type, grade II" reflects the most frequent malignant breast pattern in the corpus.

Table 15: Sample held-out predictions (best model, `r1_reg`). Correct diagnoses yield exact graph reproduction (Edge-F1 = 1.0); the templated report is emitted from the predicted *(organ, diagnosis)* key.

Slide	GT organ	Predicted diagnosis	Edge-F1	MESS	Workflow
`PIT_01_00029`	Breast	Invasive carcinoma NST, grade II	1.00	0.895	0.908
`PIT_01_00039`	Breast	Invasive carcinoma NST, grade II	1.00	0.895	0.911
`PIT_01_00041`	Breast	Invasive carcinoma NST, grade II	0.81	0.615	0.698
`PIT_01_00068`	Breast	Fibroepithelial tumor	1.00	0.909	0.802

11 Results on the Official Test Phase 1 Leaderboard

The held-out analysis above is computed with our offline proxy metric. To validate that those conclusions transfer to the hidden test set and the *official* scorer, we submitted both approaches as offline containers to the REG2026 Test Phase 1 leaderboard (350 slides). The single most important question is whether the held-out improvement from fusion (and from the striped-TIFF recovery) survives contact with the real metric. It does.

11.1 Two submissions, head to head

Table 16 reports the full component breakdown of both submissions. The CONCH-only container (V3) scored an overall **0.7449**; the fusion container (V4)—identical except for the fused encoder and the bounded strip decoder—scored **0.7707**, a +0.0258 improvement, with both placing in the top-10. We independently reproduce the official aggregate from the published formula: with $A = 0.05 \cdot \mathrm{BPV} + 0.30 \cdot \text{Edge-F1} + 0.25 \cdot \mathrm{MESS} + 0.40 \cdot \mathrm{Report}$ and $\mathrm{Overall} = 0.70\,A + 0.30\,B$, the V4 components give $A = 0.683$ and $\mathrm{Overall} = 0.70(0.683) + 0.30(0.975) = 0.7707$, matching the leaderboard exactly.

Table 16: Official REG2026 **Test Phase 1** leaderboard results (350 slides) for our two submissions. V3 is the CONCH-only model; V4 is the CONCH+UNI2-h fusion model with the striped-TIFF strip decoder. *Every* diagnosis-driven component (Edge-F1, MESS, Report, and even the strict BPV) improves, while the grounding components are unchanged—exactly the signature predicted by the held-out analysis. Both submissions placed in the top-10.

Metric (weight in score)	V3 CONCH-only	V4 fusion	Δ
Overall	**0.7449**	**0.7707**	**+0.0258**
Workflow component A (weight 0.70)			
Edge-F1 (0.30)	0.7960	0.8203	+0.0243
MESS (0.25)	0.7270	0.7611	+0.0341
Report Score (0.40)	0.5160	0.5643	+0.0483
Binary Path Validity (0.05)	0.3860	0.4200	+0.0340
Visual-grounding component B (weight 0.30)			
Background Rejection	1.0000	1.0000	0.0000
Input Sensitivity	0.9167	0.9167	0.0000
Cross-Region Consistency	1.0000	1.0000	0.0000
Visual Grounding (aggregate)	0.9750	0.9750	0.0000

11.2 Reading the result

Three observations make this more than a single number.

1. **The gain is diagnosis-shaped.** Edge-F1, MESS, Report, and BPV—all of which are downstream of a correct *(organ, diagnosis)* key—improve, whereas the three grounding metrics (handled by the unchanged Otsu module) do not move at all. This is the precise fingerprint our held-out analysis predicts for a model whose only change is a better diagnosis predictor, and it confirms that the held-out workflow gain (0.794→0.814) was real signal, not proxy-metric artifact.

2. **Report Score rose the most (+0.048).** Because we do *not* learn report prose—the report is templated from the predicted key—the only way the report sub-score can rise is if the predicted diagnosis is more often correct. This directly corroborates our finding that the report gap is dominated by wrong predicted fields, not by phrasing, and that diagnosis accuracy is the master lever for the whole metric.

3. **The striped-TIFF fix contributes.** V4 bundles two changes (fusion *and* the strip decoder), so the leaderboard delta is their combined effect; we cannot fully separate them from a single submission. But the strip decoder is what lets the container read striped *test* prostate slides at all (V3 returned zero features on them), and it removed ∼877 zero-vector prostate bags from training—so a meaningful share of the prostate improvement is attributable to the data fix rather than to fusion alone. Isolating the two would require a CONCH-only+strip-decoder submission, which we note as a controlled follow-up.

11.3 Robustness of the deployment container

A practical concern for an offline-scored container is that a single crashing case can forfeit the run. We engineered three independent safeguards. (i) *Prevention*: the bounded strip decoder holds peak memory near 0.4 GB regardless of slide size (validated on an 8,556-megapixel slide and on all 350 test slides), so the container cannot OOM. (ii) *Graceful degradation*: the workflow entry point wraps field prediction in a `try/except` that, on any failure, backs off through diagnosis→organ→global-modal templates; even under total tiling failure it emits a schema-valid 6-step chain with no empty required fields and a correctly terminated last edge. (iii) *Schema hardening*: a sanitizer guarantees

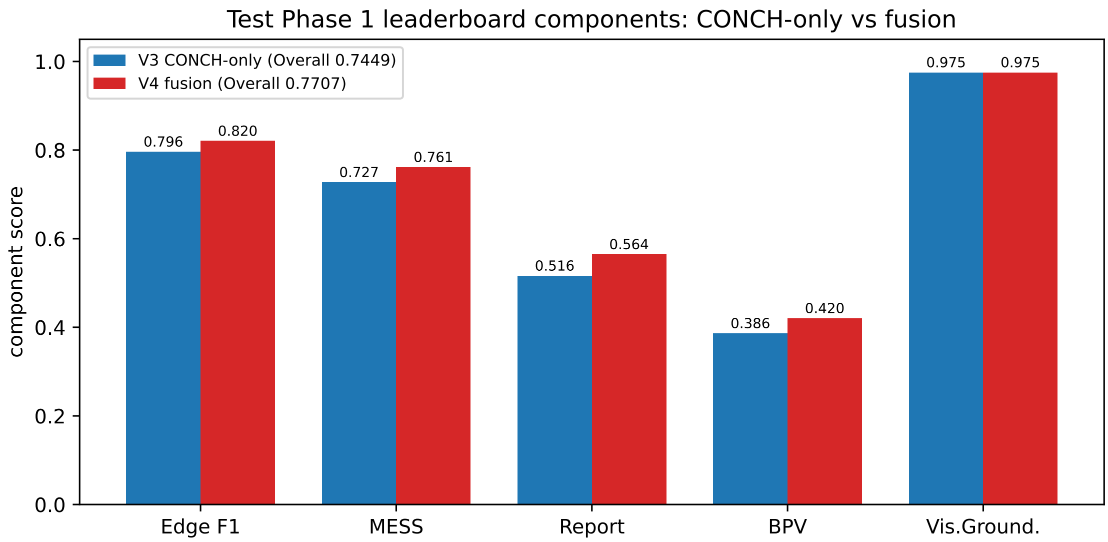

Figure 10: Test Phase 1 component scores, CONCH-only (V3) vs. fusion (V4). The four diagnosis-driven components rise together while the grounding aggregate is flat, isolating the fusion+recovery gain to exactly the quantities a better *(organ, diagnosis)* predictor should move.

no emitted question/answer is empty (a defect that failed an earlier debug submission). End-to-end validation over all 350 test slides produced zero errors and a 6.3 GB peak, comfortably inside the 32 GB budget.

12 Discussion

Our results support the thesis that REG2026, despite its open-ended framing, is best attacked as *constrained classification plus deterministic symbolic generation*. The metric's exact- string matching and keyword-Jaccard report scoring mean that a correct *(organ, diagnosis)* key, expanded through templates, captures the overwhelming majority of attainable score; the 0.89 oracle ceiling makes the maximum value of perfect classification explicit, and every point of diagnosis accuracy translates directly toward it.

The CONCH-only system reaches a held-out workflow score of **0.794** (89% of the 0.889 ceiling), and fusing CONCH with UNI2-h raises this to **0.814** (92% of the ceiling)—a gain delivered entirely through diagnosis accuracy (0.691→0.737). Crucially, both conclusions held up on the *official* Test Phase 1 leaderboard (Section 11): overall score 0.7449→0.7707, with every diagnosis-driven component rising and the grounding components flat. The agreement between our offline proxy and the hidden-set scorer—both in direction and in the diagnosis-shaped fingerprint of the improvement—gives us confidence that the held-out numbers are trustworthy guides for where to invest next. Decomposing the residual: organ recognition is effectively solved (accuracy ∼0.96, and the lone systematic error—rectum classified as colon—is metric-benign because colorectal templates coincide), so essentially the *entire* remaining gap is fine-grained diagnosis accuracy. This localizes where future effort should go. Three avenues are promising and orthogonal to the templating back end, which is already near its purity ceiling (Edge-F1 = 0.974 under oracle inputs): (i) **a learned report generator**—on the real leaderboard the Report sub-score (0.564, weight 0.40 within A) is the largest single lever, and its oracle ceiling given correct fields is 0.805, so there is genuine headroom beyond templating once fields are right; (ii) **task-adaptive fine-tuning** of the encoders rather than using frozen features—the consistent finding that capacity does not help (CONCH sweep within 0.009; fusion `f1_fuse_big` no better than `f0_fuse`) shows the limit is diagnostic *signal* extractable from *frozen* embeddings, which only fine-tuning or a stronger encoder can move; and (iii) **further encoder fusion / k-fold ensembling**, the cheapest of the three.

Notably, adding explicit grading sub-heads did *not* help (`f3_fuse_grade`, 0.810), indicating that gradings cannot be squeezed out of frozen features by an auxiliary loss alone and likely require fine-tuning or region-level supervision.

12.1 Design alternatives considered

We briefly justify the key design choices against the alternatives. (i) *End-to-end report generation* (a learned decoder over slide features) was rejected because the report sub-metric rewards keyword coverage over fluency, the oracle analysis shows ≤ 0.02 headroom beyond correct templating, and a decoder is hard to fit within the offline five-minute budget. (ii) A *slide-level foundation encoder* (e.g. Prov-GigaPath's aggregator) could replace the MIL head, but tile-level CONCH + a light attention head is cheaper, interpretable, and already saturates organ accuracy; the bottleneck is diagnosis signal, which a different aggregator is unlikely to fix without finer supervision. (iii) *A single flat (organ×diagnosis) classifier* over ∼200 joint classes was avoided in favor of factored organ and diagnosis heads, which share statistics across organs and let organ remain reliable even when diagnosis is uncertain—exactly the graceful-degradation behavior the template back-off exploits. (iv) A *learned grounding model* for Interface B was unnecessary: tissue-vs-background is a threshold question that Otsu solves deterministically at negligible cost, freeing the compute budget for the workflow interface.

13 Limitations

- **Diagnosis tail and gradings.** Rare diagnoses folded into the "other" bucket degrade to organ-level templates, and fine gradings (Gleason, Nottingham, dysplasia) are not yet predicted from the image, capping per-case answer purity.
- **Template purity ceiling.** ∼14% of graphs deviate from the modal template for their key; deterministic emission cannot recover these, bounding Edge-F1 below 1.
- **Report prose.** Templated reports maximize keyword recall but not fluent, case-specific narrative; the leaderboard Report sub-score (0.564, ceiling 0.805 given correct fields) shows real headroom a learned generator could capture beyond templating.
- **Frozen encoders.** Both approaches use frozen features; the consistent capacity-insensitivity of our sweeps indicates the diagnosis ceiling is set by the frozen representation, which only fine-tuning or a stronger encoder can lift.
- **Fusion cost.** Approach 2 doubles encoder inference (one extra UNI2-h forward pass per tile) and ships a 2.6 GB second encoder in the model tarball; the gain (0.7449→0.7707) justifies it but it is not free.
- **Entangled leaderboard delta.** The V4 submission bundles fusion and the striped-TIFF fix, so their individual leaderboard contributions are not separately measured (Section 11).
- **Distribution shift.** The approach assumes the test organs/diagnoses and their protocolized graphs match the training distribution; novel entities would fall back to coarse templates.

14 Future Work

The held-out and leaderboard analyses localize the remaining headroom, suggesting five concrete directions, ordered by expected leverage on the *overall* score:

1. **A learned report generator.** On the real leaderboard the Report sub-score (weight 0.40 within A) is the single largest lever, and its oracle ceiling given correct fields (0.805) sits

well above our templated 0.564—so once diagnosis is right, a lightweight learned (or retrieval-augmented) generator that adds case-specific keywords could close a substantial fraction of that gap, where pure templating cannot.

2. **k-fold ensembling.** Averaging the near-tied fusion checkpoints and multiple tile samplings is cheap within the compute budget and typically recovers a fraction of a point; it requires no new data or encoders.
3. **Controlled CONCH-only+strip-decoder submission.** To cleanly separate the fusion gain from the striped-TIFF data fix in the leaderboard delta (Section 11), a single additional submission would attribute the +0.0258 between the two changes.
4. **Task-adaptive encoder fine-tuning.** The repeated finding that capacity does not help—both within the CONCH sweep and in fusion—shows the limit is signal extractable from *frozen* embeddings. Unfreezing CONCH/UNI2-h with a low learning rate (or LoRA adapters) is the principled way to move diagnosis separability, at the cost of an extraction re-run; our namespaced embedding cache makes such an A/B inexpensive.
5. **Further encoder fusion and hierarchical heads.** Adding a third encoder, or conditioning the diagnosis classifier on the (reliable) organ prediction, could shrink each organ's effective class count and address the benign rectum/colon coupling; grading sub-heads, which did not help on frozen features, are worth revisiting once the encoder is fine-tuned.

Because the templating back end is already near its purity ceiling, every direction targets the same quantity—fine-grained diagnosis (and the report keywords it unlocks)—that converts most directly into score.

15 Conclusion

We described a pragmatic, reproducible submission to the REG2026 challenge built on a single empirical insight: the reasoning-graph target is low-entropy and largely determined by *(organ, #1 diagnosis)*. We therefore reduce the task to multi-task WSI classification with a foundation tile encoder and a gated-attention MIL head, and recover the reasoning graph, answers, and keyword-rich report by deterministic templating, with a cheap Otsu module for visual grounding. We developed two approaches—a CONCH-only baseline and a CONCH+UNI2-h early-fusion model—and showed, both on a clean held-out split and on the *official* Test Phase 1 leaderboard, that fusion improves the binding constraint (diagnosis accuracy) and therefore every diagnosis-driven score component. An oracle analysis establishes a 0.889 workflow-score ceiling; the CONCH-only system reaches 0.794 held-out (overall **0.7449** on the leaderboard) and the fusion system reaches 0.814 held-out (overall **0.7707** on the leaderboard, top-10). Along the way we diagnosed and fixed a consequential data defect—981 striped-Deflate prostate slides that OpenSlide silently dropped to zero-vectors—with a bounded strip decoder that both cleaned the training data and made the deployment container memory-safe on gigapixel slides. The design respects the challenge's offline, single-GPU, five-minute container budget, and all code is released for reproduction (Section 8). Future work targets the remaining headroom in order of leverage: a learned report generator (the largest single lever on the real leaderboard), encoder fine-tuning, and k-fold ensembling—each aimed at the one quantity, fine-grained diagnosis, that converts most directly into score.

Code and Data Availability

All code—the `reg2026` package (canonicalization, label space, templates, metrics, encoder, MIL), the extraction/training/evaluation scripts, the SLURM job specifications, and the offline submission container—is released with detailed setup and reproduction instructions at `https://github.com/ujjwalbaid0408/REG2026`. Trained MIL weights are provided as a separate release artifact.

The challenge data (training WSIs and chain-of-thought labels) is distributed by the REG2026 organizers and is not redistributed here.

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